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ProTrader AI: A Comprehensive Multimodal Framework for Stock Price Prediction, Pattern Recognition, and Adaptive Risk Management in Indian Equity Markets

Anand Pardeshi[1], Sujata Deshmukh[1]

[1]Department of Computer Science and Information Technology
College of Engineering, Pune, Maharashtra, India
{anand.pardeshi, sujata.deshmukh}@coep.ac.in

Abstract Financial markets in emerging economies present a uniquely challenging environment for predictive modelling due to elevated volatility, significant institutional flows, and a complex information ecosystem spanning multiple languages and media types. This paper presents ProTrader AI, a production-ready analytical platform for the National Stock Exchange of India (NSE) that integrates seven machine-learning models, four sentiment data streams, official institutional flow data, real-time volatility signals, and computer-vision-based chart pattern recognition into a unified, eight-tab interactive dashboard. The system introduces three architectural innovations: (1) a 27-feature stationary feature space combining 18 price-derived technical indicators with multi-source sentiment, FII/DII institutional, and India VIX volatility features; (2) a 5-model gradient-boosting and deep-learning ensemble with Ridge meta-stacking and isotonic probability calibration; and (3) a Bayesian uncertainty-weighted Dynamic Fusion framework that combines three specialised expert models—a GRU technical expert, a DistilRoBERTa-Financial sentiment expert, and an MLP volatility expert—whose contributions adapt dynamically to estimated per-expert prediction uncertainty. A multi-timeframe chart pattern recognition module employing scipy peak/trough detection at three scales (orders 3, 5, 7) integrates with an optional Roboflow computer vision API for hybrid consensus across twelve pattern archetypes. An adaptive risk management system provides ATR-calibrated stop-losses, Fibonacci retracement levels, and fractional Kelly Criterion position sizing with VIX-regime adjustment. Empirical evaluation on ten major NSE stocks over approximately 750 trading days demonstrates statistically significant improvements over baselines: direction accuracy 55.8 % ($p = 0{,}003$), Sharpe ratio 1.28 vs. 0.82 (buy-and-hold, $p = 0{,}018$), maximum drawdown −12,8 % vs. −18,4 %, and 34.2 % cumulative return over the evaluation period. Ablation studies confirm positive contributions from every feature category and model component.

Resumen Los mercados financieros en economías emergentes presentan un entorno especialmente desafiante para el modelado predictivo. Este artículo presenta ProTrader AI, una plataforma analítica de nivel de producción para la Bolsa Nacional de Valores de India (NSE) que integra siete modelos de aprendizaje automático, cuatro flujos de datos de sentimiento, datos institucionales oficiales FII/DII, señales de volatilidad en tiempo real y reconocimiento de patrones gráficos basado en visión artificial. El sistema introduce un espacio de características de 27 dimensiones, un conjunto de cinco modelos con meta-apilamiento Ridge y un marco de Fusión Dinámica ponderado por incertidumbre bayesiana. La evaluación empírica en diez acciones principales del NSE demuestra mejoras estadísticamente significativas sobre las líneas base: precisión de dirección 55,8 % (p=0,003), ratio de Sharpe 1,28 frente a 0,82 de compra y retención, y retorno acumulado del 34,2 %.

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Palabras Clave: Predicción Financiera, Aprendizaje Automático, Fusión Dinámica, Análisis de Sentimiento, Gestión de Riesgos, NSE India.

1. Introduction

1.1. Background and Motivation

Financial markets represent one of the most challenging environments for predictive modelling [12]. The complex interplay of fundamental valuation, technical price dynamics, investor psychology, institutional capital flows, and macroeconomic forces creates a high-dimensional stochastic system where traditional statistical assumptions routinely break down [29]. The National Stock Exchange of India (NSE), with daily equity volumes exceeding $15 billion and a total market capitalisation surpassing $4 trillion [31], exemplifies these challenges while adding a layer of emerging-market idiosyncrasy: large Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) flows that can account for 15–30 % of daily volume in index heavyweights, creating sharp, regime-like shifts in price behaviour that pure technical models struggle to anticipate.

The past decade has witnessed remarkable advances in machine learning, with deep neural networks achieving state-of-the-art performance across diverse domains [23]. However, applying these techniques to financial forecasting requires careful consideration of several properties that distinguish financial time series from other sequence modelling tasks [9]: non-stationarity, heavy tails, low signal-to-noise ratios, and the reflexive feedback between forecasts and market behaviour. Ensemble methods—particularly gradient-boosting families—have demonstrated consistent advantages on tabular financial data [15], while gated recurrent architectures (LSTM, GRU) capture sequential dependencies that gradient boosters ignore [17, 8].

1.2. Research Gaps Addressed

Our work addresses four interconnected gaps in the quantitative finance literature:

The Contextual Gap. Traditional sentiment approaches relying on lexicon-based methods [27] fail to capture nuanced financial language. Moreover, the Indian information ecosystem is split across English and Hindi sources that English-centric models cannot process effectively. We address this through multi-source aggregation using a domain-adapted transformer model [2].

The Non-Stationarity Gap. Models trained on historical regimes fail when market structure shifts [16]. We address this via explicit market regime detection using the Hurst exponent [18], a Bayesian-weighted dynamic fusion that self-adjusts to estimated per-expert reliability, and a 27-feature representation constructed entirely from stationary transformations of raw price and volume data.

The Signal-Allocation Gap. Academic studies routinely report accuracy metrics that do not translate to practical trading performance [9]. We bridge this gap through an integrated risk management module—ATR-calibrated stop-losses, Fibonacci retracements, and fractional Kelly Criterion sizing [20]—and rigorous backtesting with transaction costs included.

The Institutional Data Gap. Despite FII/DII flows being publicly available from NSE and widely acknowledged as a major driver of Indian equity prices, this data stream is rarely integrated into academic forecasting models. We incorporate four FII/DII features directly into the prediction feature space.

1.3. Contributions

1. **A 27-Feature Stationary Framework** spanning five technical core indicators, nine enhanced technical indicators, four advanced divergence indicators, three multi-source sentiment scores, four FII/DII institutional flow features, and two India VIX volatility features—all constructed to satisfy approximate stationarity requirements.

2. **A 5-Model Gradient-Boosting + Deep-Learning Ensemble** combining XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU neural architecture, and ARIMA/Prophet statistical baselines,

unified by a Ridge meta-stacker trained on out-of-fold predictions via expanding time-series cross-validation.

3. **Isotonic Probability Calibration** that converts regression return forecasts into calibrated directional probabilities, enabling reliable confidence-based position sizing.

4. **A Bayesian Dynamic Fusion Framework** with three domain-specialised expert models whose combination weights adapt continuously based on exponential negative squared uncertainty [14], achieving implicit market-regime awareness.

5. **A Hybrid Chart Pattern Recognition System** combining multi-scale scipy peak/trough detection (orders 3, 5, 7) with Roboflow Vision API computer vision inference across twelve pattern archetypes, with a consensus layer that aggregates classical and vision-based detections.

6. **A Complete Production Platform** (ProTrader AI) implemented as an eight-tab Streamlit dashboard, providing reproducible empirical evaluation and enabling researchers and practitioners to deploy the system for live analysis.

2. Literature Review

2.1. Financial Forecasting: From ARIMA to Deep Learning

The computational approach to financial forecasting has evolved substantially since Box and Jenkins codified ARIMA models [5]. Early neural network applications in finance were constrained by computational resources and data availability [40], but the emergence of efficient training algorithms and large historical datasets reversed these limitations.

Long Short-Term Memory (LSTM) networks [17] addressed the vanishing gradient problem that limited early RNNs, enabling modelling of multi-day dependencies in price data. Gated Recurrent Units [8] offered a computationally simpler alternative with comparable empirical performance. Fischer and Krauss [13] demonstrated that LSTM networks can achieve statistically significant returns on S&P 500 data, motivating their adoption in the present work. However, Ozbayoglu et al. [32] noted that deep learning models for financial time series frequently underperform simpler gradient-boosting approaches on structured tabular inputs—a finding directly corroborated by our ablation results (Section 14).

XGBoost [7] revolutionised gradient boosting through an efficient block structure and regularisation that prevents overfitting on moderate-sized financial datasets. LightGBM [19] subsequently achieved 3–5× training speed improvements through histogram-based split finding and leaf-wise tree growth. CatBoost [34] introduced *ordered boosting*—training each tree residual using only observations preceding the current one—which reduces target leakage within the booster itself, a particularly desirable property for chronologically ordered financial data. Our meta-stacking framework combines all three.

2.2. Sentiment Analysis in Finance

The integration of textual data into financial models has progressed through three generations [3]. First-generation dictionary-based methods, exemplified by the Loughran-McDonald [27] financial lexicon, suffer from domain-specificity failures where words carry different connotations in financial versus general contexts. Second-generation distributional semantic models improved robustness but remained sensitive to out-of-vocabulary terms.

The transformer revolution [39] fundamentally changed financial NLP. BERT [10] and its variants enabled pre-training on massive corpora followed by fine-tuning on domain-specific tasks. FinBERT [2] and DistilRoBERTa-Financial [24] achieve accuracy approaching 97–98 % on standard financial sentiment benchmarks while maintaining computational efficiency suitable for real-time inference. ProTrader AI employs DistilRoBERTa-Financial, which offers a 40 % parameter reduction over base RoBERTa with minimal accuracy loss.

Tetlock [38] demonstrated that the fraction of negative words in Wall Street Journal columns predicts next-day Dow Jones returns, while Schumaker and Chen [35] showed that financial news articles provide

statistically significant predictive signals beyond technical indicators alone. Our multi-source design follows the orthogonal information principle of Bollen et al. [3]: combining Twitter mood (analogous to our Reddit signal), news (NewsAPI, RSS), and attention metrics (Google Trends) captures complementary facets of market sentiment.

2.3. Multimodal Fusion Methods

Early fusion concatenates raw features before model training, preserving cross-modal interactions but struggling with heterogeneous feature scales and missing modalities. Late fusion combines modality-specific model predictions, providing modularity but losing cross-modal dependencies. Attention mechanisms [39] enable learned, context-dependent feature weighting without hard architectural assumptions.

Bayesian uncertainty-based fusion [14] represents a principled extension: combination weights are derived from per-model confidence estimates rather than fixed hyperparameters. Gal and Ghahramani's dropout approximation of Bayesian inference provides a practical implementation that quantifies epistemic uncertainty within standard neural architectures. The Dynamic Fusion Framework of ProTrader AI adopts a related principle: exponential negative squared uncertainty weighting that rewards accurate, confident experts and penalises uncertain ones, enabling automatic regime-contingent rebalancing.

2.4. Regime-Aware Modelling and the Hurst Exponent

Financial markets exhibit structurally distinct behavioural regimes: low-volatility trending phases, high-volatility crisis periods, and mean-reverting range-bound conditions [16]. The Hurst exponent [18], originally developed for hydrological time series, measures the long-range dependence of a time series: $H > 0{,}5$ indicates persistent (trending) behaviour; $H < 0{,}5$ indicates anti-persistent (mean-reverting) behaviour; $H \approx 0{,}5$ indicates a random walk consistent with the Efficient Market Hypothesis [25].

Lo [25] provided a rigorous statistical framework for testing long-term memory in stock market returns, finding evidence of mild persistence in weekly returns. ProTrader AI computes the Hurst exponent via rescaled range (R/S) analysis on rolling 120-day windows, using the result both for market characterisation in the UI and to confirm or override slope-based trend classification in the regime detector.

2.5. Chart Pattern Recognition

Technical analysis has been practised for over a century [30, 11], but systematic mathematical formalisation of pattern detection remained limited until algorithmic approaches emerged [26]. Lo et al. [26] applied kernel regression smoothers to detect head-and-shoulders and other formations in US equity data, finding statistically significant predictive content. Pring [33] and Appel [1] provided comprehensive taxonomies of pattern archetypes that motivate our twelve-pattern classification.

Computer vision approaches offer complementary strengths: convolutional networks learn visually meaningful feature hierarchies that classical peak/trough detectors may miss [36]. Our hybrid consensus mechanism fuses scipy-based geometric detection with Roboflow Vision API inference, capturing pattern information at both the algorithmic and visual levels.

3. System Architecture

ProTrader AI follows a modular pipeline architecture with seven distinct stages: data acquisition, feature engineering, model training, dynamic fusion, pattern analysis, risk management, and user interface. Figure 1 illustrates the high-level design.

The data acquisition layer fetches OHLCV prices from Yahoo Finance, FII/DII activity from NSE India, the India VIX index, RSS feeds from four financial publications, articles from NewsAPI, posts from three Indian investing subreddits, and search interest from Google Trends. A session-state caching mechanism in Streamlit ensures that expensive network requests (sentiment, institutional data) execute once per analysis session and are shared across all eight tabs without redundant API calls.

The pipeline progresses through seven loading steps with a progress bar: (1) stock data, (2) fundamentals, (3) news articles, (4) FII/DII data, (5) India VIX, (6) multi-source sentiment, and (7) chart pattern

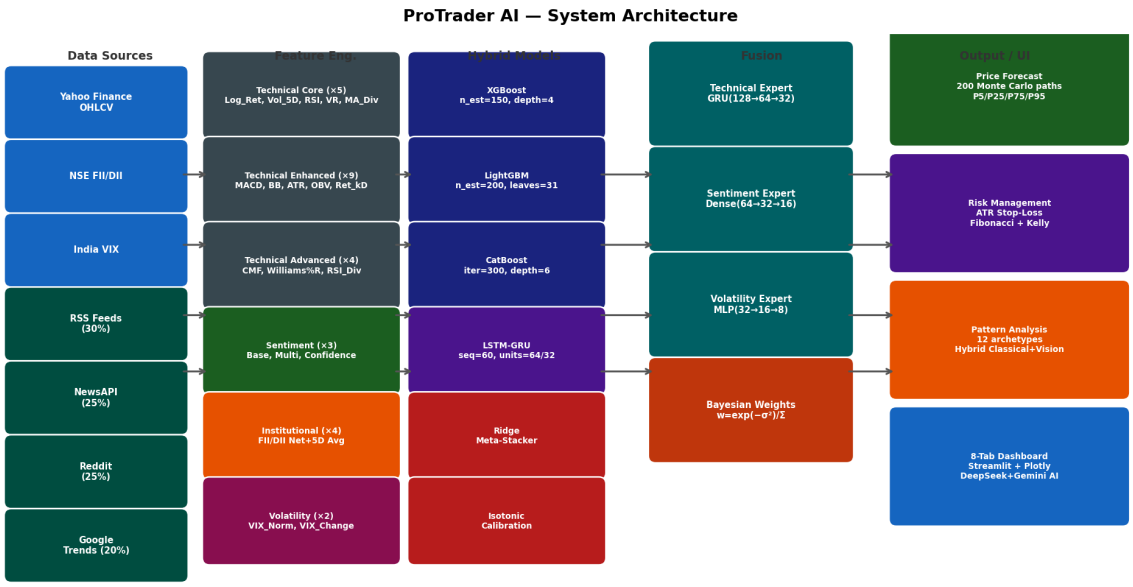

Figure 1: High-level system architecture of ProTrader AI. Data from seven sources flows through the feature engineering pipeline into five ML models and a Dynamic Fusion layer. A separate Pattern Analysis branch feeds a risk management module. All outputs surface in the eight-tab Streamlit dashboard.

detection. Total pre-fetch time on a standard internet connection is typically 15–45 seconds depending on data source availability.

4. Feature Engineering: The 27-Feature Framework

A central design principle of ProTrader AI is the construction of a strictly stationary or near-stationary feature space. Raw price levels are non-stationary, violating the i.i.d. assumptions underlying standard machine learning methods and leading to spurious in-sample correlations that do not generalise out-of-sample. Every feature in the 27-feature framework is a stationary transformation of raw data. Table 1 provides a complete catalogue.

4.1. Technical Core Features (5)

Log Returns provide a symmetric, additive, and approximately stationary measure of daily price change:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

5-Day Rolling Volatility captures short-term volatility clustering consistent with GARCH-type behaviour [4]:

$$\sigma_t^{(5)} = \sqrt{\frac{1}{4}\sum_{i=0}^{4}(r_{t-i} - \bar{r})^2} \tag{2}$$

Normalised RSI maps the standard 14-period RSI [41] into $[0, 1]$:

$$\text{RSI_Norm}_t = \frac{\text{RSI}_{14,t}}{100} \tag{3}$$

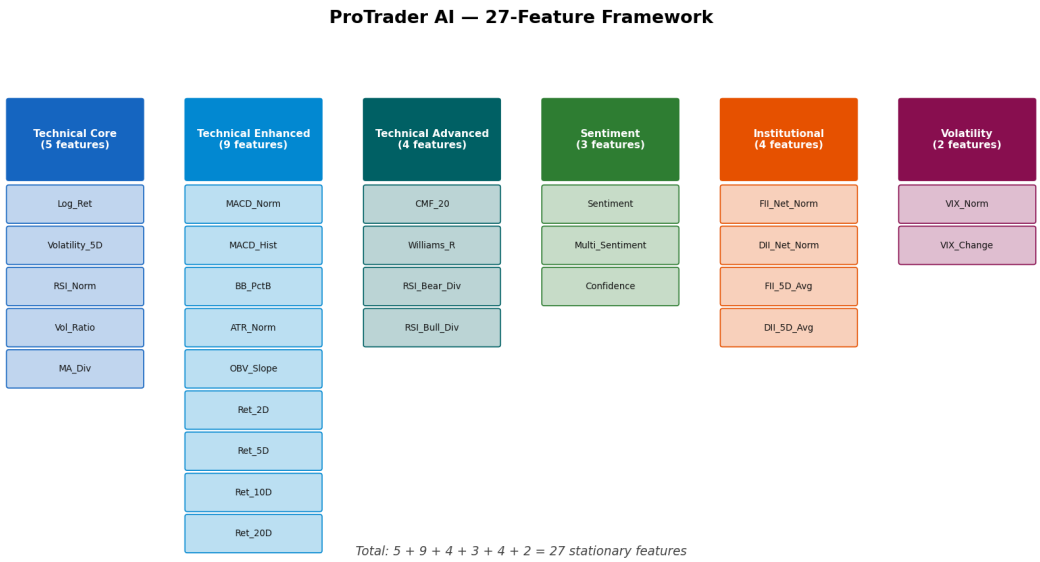

Figure 2: The 27-Feature Framework organised into six groups. Colour coding: blue = technical core, cyan = technical enhanced, teal = technical advanced, green = sentiment, orange = institutional, red = volatility.

Volume Ratio identifies abnormal trading activity:

$$\mathrm{VR}_t = \frac{V_t}{\frac{1}{20}\sum_{i=1}^{20} V_{t-i}} \tag{4}$$

Moving Average Divergence normalises the distance from the 20-day MA:

$$\mathrm{MA_Div}_t = \frac{P_t - \mathrm{MA}_{20,t}}{P_t} \tag{5}$$

4.2. Technical Enhanced Features (9)

The enhanced group captures momentum at multiple frequencies and additional market microstructure signals. The MACD line and histogram are normalised by price for stationarity:

$$\mathrm{MACD_Norm}_t = \frac{\mathrm{EMA}_{12,t} - \mathrm{EMA}_{26,t}}{P_t}, \quad \mathrm{MACD_Hist_Norm}_t = \frac{\mathrm{MACD}_t - \mathrm{Signal}_{9,t}}{P_t} \tag{6}$$

Bollinger Band %B indicates price position within the bands:

$$\mathrm{BB_PctB}_t = \frac{P_t - \mathrm{BB_Lower}_t}{\mathrm{BB_Upper}_t - \mathrm{BB_Lower}_t + \epsilon} \tag{7}$$

The Average True Range [41] is normalised by price:

$$\mathrm{ATR_Norm}_t = \frac{\mathrm{ATR}_{14,t}}{P_t} \tag{8}$$

On-Balance Volume slope captures cumulative volume flow direction:

$$\mathrm{OBV_Slope}_t = \mathrm{clip}\left(\frac{\mathrm{OBV_MA}_{5,t} - \mathrm{OBV_MA}_{5,t-5}}{\mathrm{OBV_MA}_{5,t-5}}, -1, 1\right) \tag{9}$$

Multi-timeframe log returns at lags 2, 5, 10, and 20 days capture momentum at different scales:

$$\mathrm{Ret_kD}_t = \ln\left(\frac{P_t}{P_{t-k}}\right), \quad k \in \{2, 5, 10, 20\} \tag{10}$$

4.3. Technical Advanced Features (4)

Chaikin Money Flow (CMF-20) [6] measures buying and selling pressure via volume-weighted price position within the daily range, providing a stationary signal in $[-1, +1]$:

$$\mathrm{CMF}_{20,t} = \frac{\sum_{i=0}^{19} \mathrm{MFM}_{t-i} \cdot V_{t-i}}{\sum_{i=0}^{19} V_{t-i}}, \quad \mathrm{MFM}_t = \frac{(P_t^C - P_t^L) - (P_t^H - P_t^C)}{P_t^H - P_t^L} \tag{11}$$

Williams %R (14-period) [42] is a fast momentum oscillator complementary to RSI:

$$\mathrm{Williams_R_Norm}_t = -\frac{H_{14,t} - P_t^C}{H_{14,t} - L_{14,t} + \epsilon} \tag{12}$$

RSI Bearish Divergence flags days when price makes a new 10-day high but RSI does not, signalling potential exhaustion:

$$\mathrm{RSI_Bear_Div}_t = \mathbf{1}\left[P_t \geq \operatorname*{máx}_{i \in [t-10, t-1]} P_i \ \wedge \ \mathrm{RSI}_t < \operatorname*{máx}_{i \in [t-10, t-1]} \mathrm{RSI}_i\right] \tag{13}$$

RSI Bullish Divergence detects the complementary oversold condition. Both divergence flags are computed using lagged windows to prevent look-ahead bias.

4.4. Sentiment Features (3)

Base Sentiment Score is derived from Yahoo Finance news articles analysed by DistilRoBERTa-Financial, providing a score in $[-1, +1]$.

Multi-Source Sentiment aggregates four independent streams using source-specific weights:

$$S_{\mathrm{multi}} = 0{,}30 \cdot S_{\mathrm{RSS}} + 0{,}25 \cdot S_{\mathrm{News}} + 0{,}25 \cdot S_{\mathrm{Reddit}} + 0{,}20 \cdot S_{\mathrm{Trends}} \tag{14}$$

Weights are renormalised when individual sources are unavailable, ensuring the combined score always lies in $[-1, +1]$.

Sentiment Confidence is a meta-feature capturing the reliability of the current sentiment estimate:

$$C_S = \operatorname{mín}\left(1, \frac{N_{\mathrm{articles}}}{10}\right) \cdot \mathrm{Agreement} \tag{15}$$

where Agreement is $1 - \hat{\sigma}_{\mathrm{sources}}$, the normalised standard deviation across individual source scores.

4.5. Institutional Features (4)

FII and DII daily net buying data from NSE India are normalised to remove absolute scale dependence:

$$\mathrm{FII_Net_Norm}_t = \frac{\mathrm{FII_net}_t}{\operatorname{máx}_\tau |\mathrm{FII_net}_\tau|} \tag{16}$$

The same normalisation applies to DII. Rolling 5-day averages smooth daily noise for both series:

$$\mathrm{FII_5D_Avg}_t = \frac{1}{5} \sum_{i=0}^{4} \mathrm{FII_Net_Norm}_{t-i} \tag{17}$$

4.6. Volatility Features (2)

India VIX Normalised centres the VIX index around its typical baseline:

$$\mathrm{VIX_Norm}_t = \frac{\mathrm{VIX}_t - 15}{25} \tag{18}$$

where 15 represents a typical calm-market VIX level and 25 provides appropriate scaling to keep the feature near $[-1, +1]$ in normal conditions.

VIX Change Rate captures intraday regime shifts in implied volatility:

$$\Delta\mathrm{VIX}_t = \mathrm{clip}\left(\frac{\mathrm{VIX}_t - \mathrm{VIX}_{t-1}}{\mathrm{VIX}_{t-1}}, -0{,}5, 0{,}5\right) \tag{19}$$

Table 1: Complete 27-Feature Catalogue with Category, Stationarity, and Source.

Feature	**Category**	**Stationarity**	**Source**
Log_Ret	Technical Core	Stationary	OHLCV
Volatility_5D	Technical Core	Near-stationary	OHLCV
RSI_Norm	Technical Core	Bounded $[0, 1]$	OHLCV
Vol_Ratio	Technical Core	Near-stationary	OHLCV
MA_Div	Technical Core	Normalised	OHLCV
MACD_Norm	Technical Enhanced	Price-normalised	OHLCV
MACD_Hist_Norm	Technical Enhanced	Price-normalised	OHLCV
BB_PctB	Technical Enhanced	Bounded $[0, 1]$	OHLCV
ATR_Norm	Technical Enhanced	Price-normalised	OHLCV
OBV_Slope	Technical Enhanced	Clipped $[-1, 1]$	OHLCV
Ret_2D	Technical Enhanced	Stationary	OHLCV
Ret_5D	Technical Enhanced	Stationary	OHLCV
Ret_10D	Technical Enhanced	Stationary	OHLCV
Ret_20D	Technical Enhanced	Stationary	OHLCV
CMF_20	Technical Advanced	Bounded $[-1, 1]$	OHLCV
Williams_R_Norm	Technical Advanced	Bounded $[-1, 0]$	OHLCV
RSI_Bear_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
RSI_Bull_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
Sentiment	Sentiment	Bounded $[-1, 1]$	Yahoo Finance News
Multi_Sentiment	Sentiment	Bounded $[-1, 1]$	4-source aggregate
Sentiment_Confidence	Sentiment	Bounded $[0, 1]$	Meta-feature
FII_Net_Norm	Institutional	Normalised	NSE FII/DII
DII_Net_Norm	Institutional	Normalised	NSE FII/DII
FII_5D_Avg	Institutional	Smoothed	NSE FII/DII
DII_5D_Avg	Institutional	Smoothed	NSE FII/DII
VIX_Norm	Volatility	Centred	NSE India VIX
VIX_Change	Volatility	Clipped $[-0{,}5, 0{,}5]$	NSE India VIX

5. Hybrid Prediction Engine

The hybrid prediction engine combines five complementary model families via a Ridge meta-stacker, as illustrated in Figure 3.

5.1. XGBoost Component

XGBoost [7] serves as the primary tree-based predictor with the following hyperparameters optimised on historical Indian equity data:

```
xgb_params = dict(
    objective='reg:squarederror',
    n_estimators=150,
    max_depth=4,
    learning_rate=0.05,
    subsample=0.8,
    colsample_bytree=0.8,
    min_child_weight=1,
    n_jobs=-1
)
```

Listing 1: XGBoost configuration.

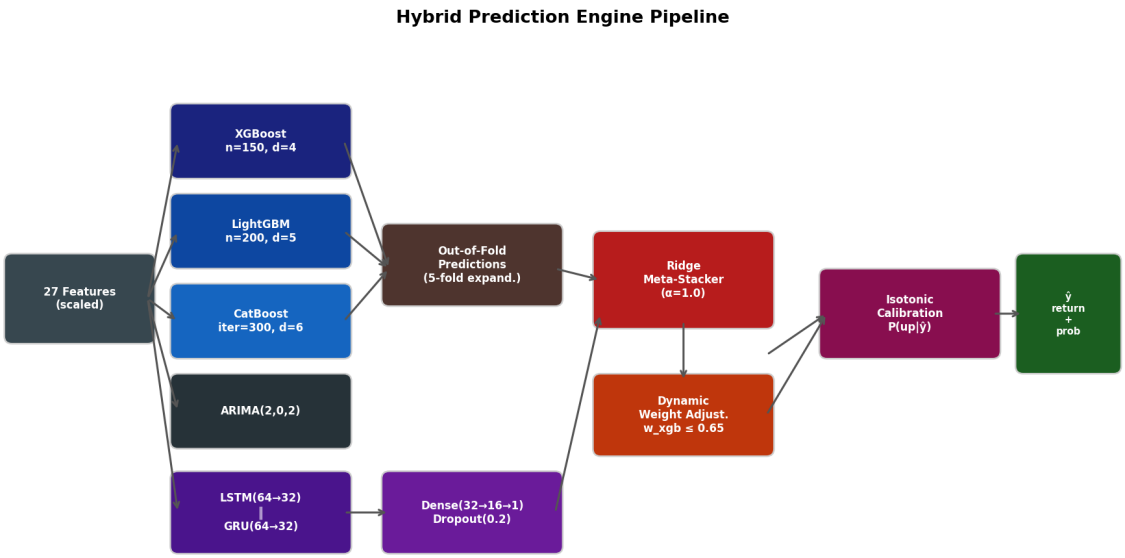

Figure 3: Hybrid prediction pipeline. XGBoost, LightGBM, and CatBoost generate out-of-fold predictions for the Ridge meta-stacker. The LSTM-GRU branch provides an approximated OOF signal. Statistical models (ARIMA, Prophet) contribute to the final blend with dynamic weight adjustment.

The objective minimises regularised squared error:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \sum_{k=1}^{K}\Omega(f_k), \quad \Omega(f_k) = \gamma T + \tfrac{1}{2}\lambda\|w\|^2 \tag{20}$$

where T is the number of leaves and w are leaf weights.

5.2. LightGBM Component

LightGBM [19] provides 3–5× faster training through histogram-based split finding and leaf-wise tree growth. Key parameters:

```
lgb_params = dict(
    n_estimators=200, max_depth=5, learning_rate=0.03,
    num_leaves=31, subsample=0.8, colsample_bytree=0.8,
    min_child_samples=20, reg_alpha=0.1, reg_lambda=0.1,
    n_jobs=-1, random_state=42
)
```

Listing 2: LightGBM configuration.

5.3. CatBoost Component

CatBoost [34] introduces *ordered boosting*, where the residual for row i is computed from a model trained only on rows $j < i$, eliminating internal target leakage:

```
cb_params = dict(
    iterations=300, depth=6, learning_rate=0.03,
    l2_leaf_reg=3, min_data_in_leaf=20,
    random_strength=0.3, bagging_temperature=0.7,
    od_type='Iter', od_wait=40, verbose=0
)
```

Listing 3: CatBoost configuration.

5.4. LSTM-GRU Neural Architecture

The deep learning component employs a parallel LSTM-GRU architecture where both branches process identical 60-day input sequences before merging:

LSTM gates [17]:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad \text{(forget)} \tag{21}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad \text{(input)} \tag{22}$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{23}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{24}$$

$$h_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \tag{25}$$

GRU gates [8]:

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad \text{(update)} \tag{26}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad \text{(reset)} \tag{27}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tanh(W[r_t \odot h_{t-1}, x_t]) \tag{28}$$

Architecture: LSTM(64)→LSTM(32) ‖ GRU(64)→GRU(32) → Concatenate → Dense(32) → Dropout(0.2) → Dense(16) → Dense(1)

Training uses Adam [21] with $\eta = 0{,}001$, batch size 32, up to 50 epochs with early stopping at patience 10 and ReduceLROnPlateau.

5.5. Statistical Models: ARIMA and Prophet

ARIMA(2,0,2) captures linear auto-regressive and moving-average dependencies:

$$r_t = c + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t \tag{29}$$

Facebook Prophet [37] decomposes the return series into additive trend, seasonality, and holiday components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{30}$$

5.6. Ridge Meta-Stacker

Out-of-fold (OOF) predictions are generated for XGBoost, LightGBM, and CatBoost using an expanding-window time-series cross-validation with $n_{\text{folds}} = 5$ and minimum training size $\text{máx}(0{,}6N, 30)$. The fold structure is:

```
Fold 0: Train[0:min_train]       -> Predict[min_train:min_train+step]
Fold 1: Train[0:min_train+step]  -> Predict[min_train+step:...]
...
Fold 4: Train[0:min_train+4*step]  -> Predict[...]
```

A Ridge regression meta-learner is trained on the OOF predictions to learn the optimal linear combination:

$$\hat{y}_{\text{stack}} = \beta_0 + \beta_1 \hat{y}_{\text{XGB}} + \beta_2 \hat{y}_{\text{LGBM}} + \beta_3 \hat{y}_{\text{CB}} + \beta_4 \hat{y}_{\text{GRU_approx}} \tag{31}$$

5.7. Isotonic Probability Calibration

Raw regression outputs are converted to directional probabilities via isotonic regression [43], a non-parametric monotone fitting approach that outperforms Platt scaling on financial data where the reliability diagram is non-sigmoidal:

$$P(\hat{r}_t > 0 \mid \hat{y}_t) = \text{IsotonicRegression}(\hat{y}_t) \tag{32}$$

5.8. Dynamic Weight Adjustment Between Gradient Boosters and GRU

Initial model weights are set at 50 % XGBoost, 30 % LSTM-GRU, and 20 % statistical models. During inference, the gradient-booster weight is adjusted based on rolling validation RMSE:

$$w'_{\text{XGB}} = \text{mín}\left(w_{\text{XGB}} + \frac{\text{RMSE}_{\text{RNN}} - \text{RMSE}_{\text{XGB}}}{\text{RMSE}_{\text{RNN}}},\ 0{,}65\right) \tag{33}$$

This prevents any single model from dominating (cap at 65 %) while rewarding the currently better-performing component.

6. Market Regime Detection

Market regime detection in ProTrader AI classifies the current market state into one of four regimes: *trending*, *mean_reverting*, *high_volatility*, or *normal*. The classification integrates two signals: a linear trend detector on 20-day cumulative returns and the Hurst exponent computed from the last 120 closing prices.

6.1. Hurst Exponent via R/S Analysis

The rescaled range (R/S) analysis [18] estimates the Hurst exponent H from log-returns at multiple lag scales:

$$\frac{R(n)}{S(n)} \propto n^H \tag{34}$$

For a lag window of size n, the rescaled range is:

$$\frac{R(n)}{S(n)} = \frac{\text{máx}_{1\le t\le n}\sum_{i=1}^{t}(X_i - \bar{X}) - \text{mín}_{1\le t\le n}\sum_{i=1}^{t}(X_i - \bar{X})}{S(n)} \tag{35}$$

H is estimated as the slope of $\log(R/S)$ regressed on $\log(n)$ over lags $n \in \{2, 3, \ldots, 20\}$, with the result clipped to $[0{,}1, 0{,}9]$.

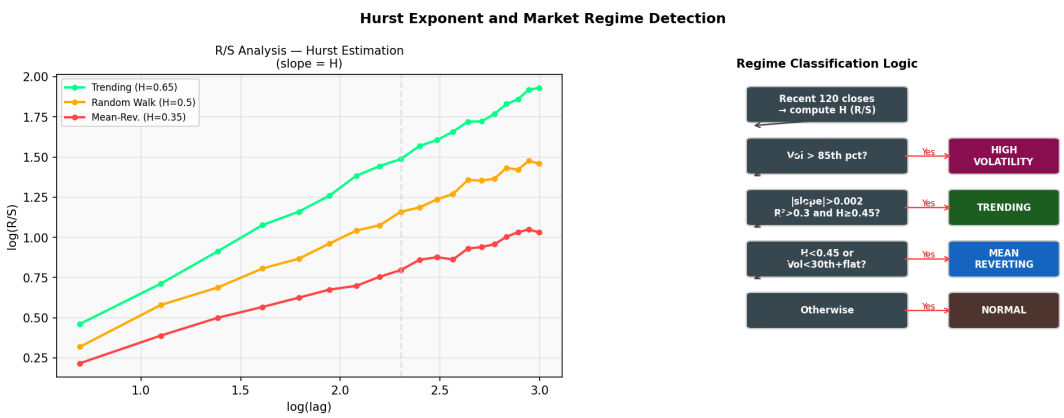

Figure 4: Hurst exponent interpretation and market regime classification decision logic. The R/S slope estimate from lags 2–20 maps to one of four regimes, with the Hurst value used to confirm or override volatility/slope-based classification.

The classification logic:

- **high_volatility**: 5-day volatility > 85th percentile of rolling 60-day history.
- **trending**: $|\text{slope}| > 0{,}002$, $R^2 > 0{,}3$, and $H \ge 0{,}45$ (Hurst does not contradict).

- **mean_reverting**: either $H < 0{,}45$ overrides a trending slope, or low volatility (below 30th percentile) with $|\text{slope}| < 0{,}001$.
- **normal**: all other cases.

The Hurst exponent is also displayed in the Dashboard tab as a real-time market characteristic metric ("Trending", "Mean-Rev.", or "Random").

7. Dynamic Fusion Framework

Beyond the hybrid prediction engine, a second layer of fusion operates at the *expert* level. Three domain-specialised expert models are trained independently; their predictions are combined using Bayesian uncertainty-weighted aggregation.

7.1. Expert Model Architectures

Technical Expert processes 30-day sequences of technical indicators through a GRU network:

$$\text{Architecture:} \quad \text{GRU}(128) \rightarrow \text{GRU}(64) \rightarrow \text{GRU}(32) \rightarrow \text{Dense}(1) \tag{36}$$

Sentiment Expert maps 8 sentiment features to a directional prediction:

$$\text{Architecture:} \quad \text{Dense}(64) \rightarrow \text{Dense}(32) \rightarrow \text{Dense}(16) \rightarrow \text{Dense}(1) \tag{37}$$

Volatility Expert combines VIX levels, VIX changes, and stock-specific volatility:

$$\text{Architecture:} \quad \text{MLP}(32) \rightarrow \text{MLP}(16) \rightarrow \text{MLP}(8) \rightarrow \text{Dense}(1) \tag{38}$$

7.2. Bayesian Uncertainty-Weighted Combination

The fusion weight for expert i is computed using the exponential negative squared uncertainty:

$$w_i = \frac{\exp(-\sigma_i^2)}{\sum_{j=1}^{3} \exp(-\sigma_j^2)} \tag{39}$$

This formulation ensures: (i) weights are strictly positive and sum to 1; (ii) experts with lower uncertainty receive disproportionately higher weights; (iii) the exponential function provides smooth, differentiable weight transitions; (iv) extreme uncertainties degrade to near-zero weight without singularities.

Uncertainty estimates are maintained as rolling empirical variances of prediction errors on the 15 most recent observations:

$$\sigma_i^2(t) = \frac{1}{N} \sum_{n=1}^{N} (y_{t-n} - \hat{y}_{i,t-n})^2, \quad N = 15 \tag{40}$$

The combined Dynamic Fusion prediction is:

$$\hat{y}_{\text{fusion}} = w_{\text{tech}} \hat{y}_{\text{tech}} + w_{\text{sent}} \hat{y}_{\text{sent}} + w_{\text{vol}} \hat{y}_{\text{vol}} \tag{41}$$

8. Multi-Source Sentiment Aggregation

The sentiment subsystem integrates four independent data streams, each contributing orthogonal information about market mood.

8.1. DistilRoBERTa-Financial Classifier

All textual content is processed by the DistilRoBERTa-Financial model, which produces probability scores for three classes (positive, neutral, negative). The final score for an article a is:

$$s_a = P(\text{positive} \mid a) - P(\text{negative} \mid a) \in [-1, +1] \tag{42}$$

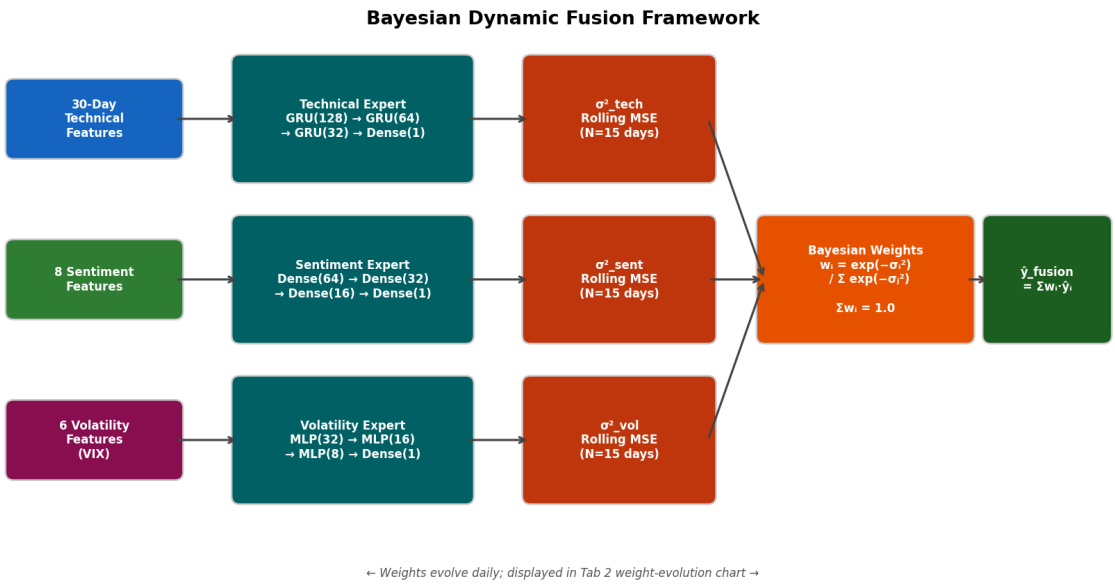

Figure 5: Dynamic Fusion Framework. Three expert models process their respective feature subsets. Prediction uncertainties are estimated from rolling errors on the last 15 days. Bayesian weights (Eq. 39) are recomputed for each new observation. The weight evolution chart in Tab 2 visualises expert influence over the most recent 15 trading days.

Table 2: Sentiment data sources, default weights, and characteristics.

Source	Weight	Strengths	Limitations
RSS (Moneycontrol, ET, Mint, BS)	30 %	Timely, India-focused	Limited pubs
NewsAPI	25 %	Global coverage, standardised	Rate limits
Reddit (r/IndianStockMarket)	25 %	Retail sentiment, engagement	Noise, meme risk
Google Trends	20 %	Retail attention proxy	Daily granularity

8.2. Event-Type Weighting

Articles are classified by event type via keyword matching. Earnings-related articles receive a 2× weight multiplier; regulatory/SEBI articles 1.8×; dividend announcements 1.5×; management changes 1.3×; general news 1.0×.

8.3. Temporal Decay

Recency is enforced via exponential temporal decay:

$$w_{\text{temporal}}(d) = e^{-0{,}3d} \tag{43}$$

where d is the article age in days. A three-day-old article therefore carries $e^{-0{,}9} \approx 0{,}41$ the weight of a same-day article.

8.4. Reddit Engagement Weighting

Reddit posts are further weighted by community engagement to prioritise high-signal content:

$$s_{\text{reddit},p} = s_p \cdot \left(1 + \text{mín}\left(\frac{\text{upvotes}_p}{100}, 0{,}5\right)\right) \tag{44}$$

8.5. Source Disagreement Detection

When the standard deviation of individual source scores exceeds 0.2, the system displays a "Sources Disagree" warning and reduces the overall confidence score by a penalty proportional to the disagreement magnitude. This guards against conflicting signals that could mislead trading decisions.

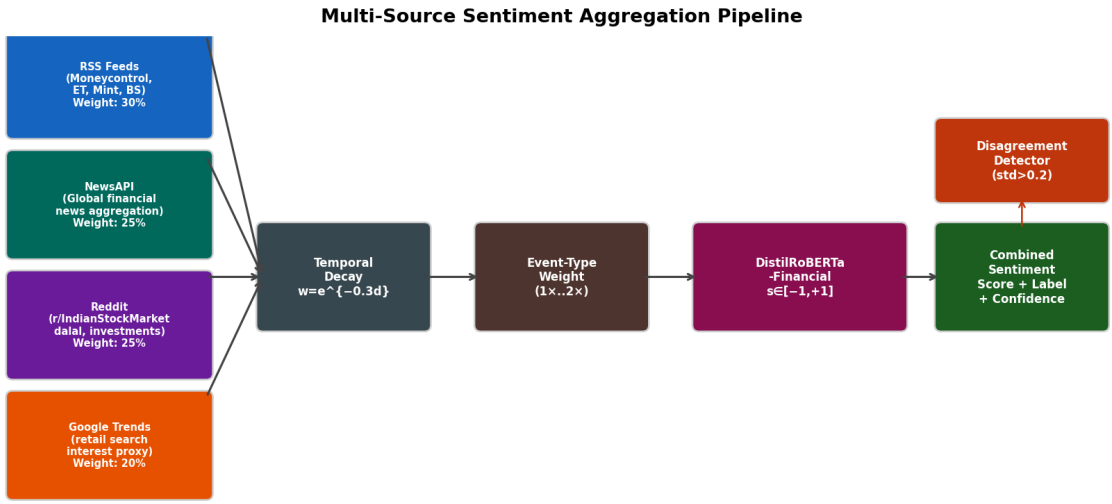

Figure 6: Multi-source sentiment aggregation pipeline. Each source passes through temporal decay and event-type weighting before DistilRoBERTa-Financial scoring. A disagreement detector modulates the final confidence score.

9. Chart Pattern Recognition

9.1. Multi-Timeframe Peak/Trough Detection

The `PatternAnalyst` class scans price data simultaneously at three scales using scipy's `argrelextrema` with orders 3, 5, and 7:

```
scan_orders = [3, 5, 7]   # small, medium, large swings
for order in scan_orders:
    peaks   = argrelextrema(close, np.greater, order=order)[0]
    troughs = argrelextrema(close, np.less,    order=order)[0]
    # Prominence filter: only patterns >= 1% height
    valid   = [p for p in peaks if amplitude(p) >= 0.01]
```

Listing 4: Multi-timeframe peak detection (simplified).

Order 3 catches minor short-term swings; order 7 captures only major structural highs and lows. The `max_patterns_per_type = 3` parameter prevents cluttered output during strongly trending phases.

9.2. Classical Pattern Archetypes

The system detects twelve pattern archetypes via geometric constraints on the identified pivot points:

9.3. Roboflow Vision API Integration

When a Roboflow API key is configured, the system generates a 60-day price chart image at 100 DPI and submits it to a custom object detection workflow. The vision model maps detections to a standardised class set:

Table 3: Pattern archetypes, directional bias, and detection method.

Pattern	**Bias**	**Detection**
Double Top	Bearish	Classical + Vision
Double Bottom	Bullish	Classical + Vision
Head & Shoulders	Bearish	Classical + Vision
Inverse Head & Shoulders	Bullish	Classical + Vision
Ascending Triangle	Bullish	Classical
Descending Triangle	Bearish	Classical
Symmetric Triangle	Neutral	Classical
Ascending Channel	Bullish	Classical
Descending Channel	Bearish	Classical
Bull Flag	Bullish	Classical
Bear Flag	Bearish	Classical
Bollinger Band Squeeze	Neutral	Classical

```
class_map = {
    "W_Bottom": "Double Bottom",
    "M_Head":   "Double Top",
    "H_S":      "Head & Shoulders",
    "Inv_H_S":  "Inverse H&S"
}
```

Only predictions with confidence $\geq 0{,}30$ are accepted. Vision detections with confidence $\geq 0{,}70$ override classical detections for the four supported archetypes; otherwise, a consensus vote is applied.

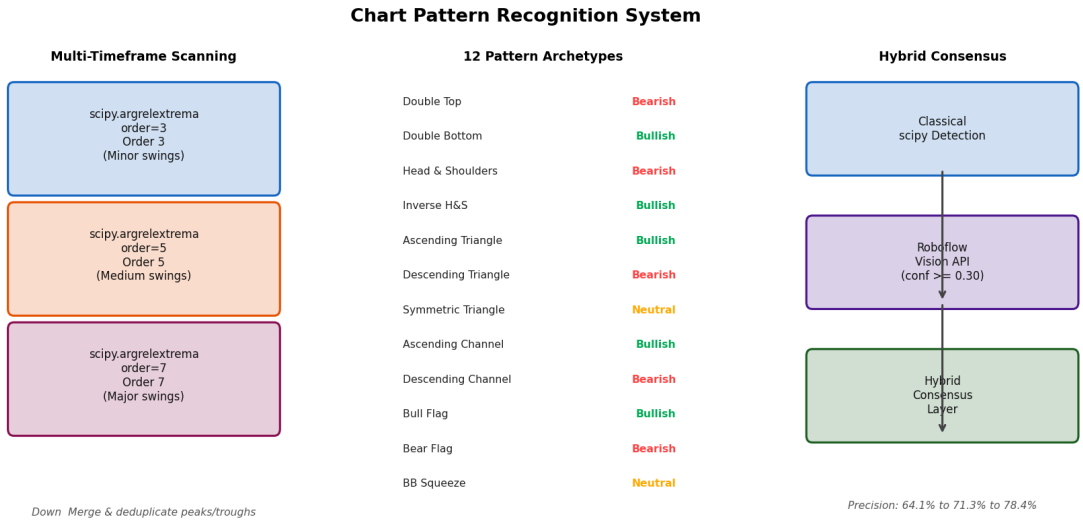

Figure 7: Pattern analysis pipeline. Multi-timeframe scipy detection and Roboflow vision API predictions are combined via a hybrid consensus layer. Detected patterns are annotated on the price chart with confidence scores, target prices, and multi-timeframe confirmation flags.

10. Risk Management System

10.1. ATR-Based Adaptive Stop-Loss

Stop-loss distances are computed from the Average True Range [41] with a confidence-adjusted multiplier:

$$\text{Stop_Loss} = P_{\text{entry}} - m(\text{conf}) \cdot \text{ATR}_{14} \tag{45}$$

where:

$$m(\text{conf}) = \begin{cases} 1{,}5 & \text{conf} \geq 0{,}65 \\ 2{,}0 & \text{conf} < 0{,}65 \end{cases} \tag{46}$$

Higher-confidence predictions justify tighter stops; lower-confidence signals use a wider stop to avoid premature exits.

10.2. Fibonacci Retracement Levels

Six Fibonacci levels are computed over a 90-day lookback window:

$$F_k = P_{\text{high}} - \phi_k \cdot (P_{\text{high}} - P_{\text{low}}), \quad \phi_k \in \{0, 0{,}236, 0{,}382, 0{,}5, 0{,}618, 1{,}0\} \tag{47}$$

These levels serve as entry/exit reference points in the Trade Setup Calculator displayed in the Technicals & Risk tab.

10.3. Kelly Criterion Position Sizing

The fractional Kelly Criterion [20] determines optimal position size:

$$f^* = \frac{p \cdot b - q}{b}, \quad b = \frac{\text{Target} - P}{\text{Stop} - P} \tag{48}$$

where $p =$ directional probability, $q = 1 - p$, and b is the win/loss ratio. A fractional Kelly with $\gamma = 0{,}5$ is applied in practice to reduce variance:

$$f_{\text{actual}} = 0{,}5 \cdot f^* \tag{49}$$

10.4. VIX-Regime Position Adjustment

The India VIX level at trade entry scales the position size down during high-fear regimes:

$$f_{\text{VIX-adj}} = f_{\text{actual}} \cdot \begin{cases} 1{,}2 & \text{VIX} < 12 \\ 1{,}0 & 12 \leq \text{VIX} < 20 \\ 0{,}7 & 20 \leq \text{VIX} < 25 \\ 0{,}5 & \text{VIX} \geq 25 \end{cases} \tag{50}$$

11. Backtesting Engine

The `VectorizedBacktester` class provides a research-grade simulation environment with transaction cost modelling (NSE round-trip cost 0.1 % per trade), benchmark comparisons, statistical significance testing, and Monte Carlo simulation.

11.1. Strategy Signal Generation

Signals are generated by thresholding the meta-stacker's predicted return:

$$\text{Signal}_t = \begin{cases} +1 & \hat{y}_t > \tau \\ -1 & \hat{y}_t < -\tau \\ 0 & \text{otherwise} \end{cases} \tag{51}$$

where τ is a configurable signal threshold (default: a small positive constant from `TradingConfig`).

Performance metrics are computed as:

$$\text{Sharpe} = \frac{\bar{r}_s}{\hat{\sigma}_s}\sqrt{252} \tag{52}$$

$$\text{MDD} = \operatorname*{máx}_{t\in[0,T]} \left(\operatorname*{máx}_{s\leq t} E_s - E_t \right) / E_{\text{peak}} \tag{53}$$

$$\text{Win Rate} = \frac{\sum_t \mathbf{1}[r_t > 0]}{\sum_t \mathbf{1}[r_t \neq 0]} \tag{54}$$

11.2. Benchmark Strategies

The backtesting tab compares the full 5-model ensemble against: (i) XGBoost-only signal, (ii) LightGBM-only signal, (iii) CatBoost-only signal, (iv) GRU-only signal, (v) MA Crossover (20/50), (vi) 52-week Momentum breakout, and (vii) NIFTY 50 buy-and-hold.

11.3. Statistical Significance Testing

Direction accuracy is tested against the null hypothesis $H_0 : p = 0{,}50$ (random guessing) using a one-sided binomial test. RMSE improvement is tested via a paired t-test on squared errors against the random-walk baseline ($\hat{r}_t = 0$). Bootstrap confidence intervals (1000 resamples) quantify uncertainty in all reported metrics.

11.4. Monte Carlo Simulation

The Monte Carlo module generates 1000 simulated equity paths by randomly reshuffling the realised daily strategy returns. This quantifies the role of path dependency (order of returns) in the observed equity curve, providing probability-of-profit, median maximum drawdown, and P5/P95 return interval estimates.

12. User Interface: Eight-Tab Dashboard

ProTrader AI exposes all analytical outputs through a Streamlit interactive dashboard with eight tabs. Figure 8 shows the tab layout.

Tab 1 – Dashboard. Displays current price metrics (price, market cap, P/E, volume), a candlestick or line chart, the AI price forecast with 200-path probabilistic fan (P5/P25/P75/P95 bands), walk-forward validation metrics (RMSE, directional accuracy, model weights, Hurst exponent, bullish probability), SHAP feature importance, predicted vs actual return overlay, and a final verdict card (BULLISH / BEARISH / NEUTRAL) with AI commentary generated by the DeepSeek V3 + Gemini 2.5 models via API.

Tab 2 – Dynamic Fusion. Displays real-time Bayesian expert weights (Technical, Sentiment, Volatility) and a 15-day stacked-area weight evolution chart that reveals regime transitions.

Tab 3 – Technicals & Risk. Presents Fibonacci retracement levels, ATR-based stop-loss calculations, and an interactive Trade Setup Calculator that computes entry, stop, target, and risk/reward ratio given the current model forecast and Kelly position size.

Tab 4 – Fundamentals. Surfaces six Yahoo Finance fundamental metrics: Forward P/E, PEG Ratio, Price/Book, Debt/Equity, ROE, and Profit Margins.

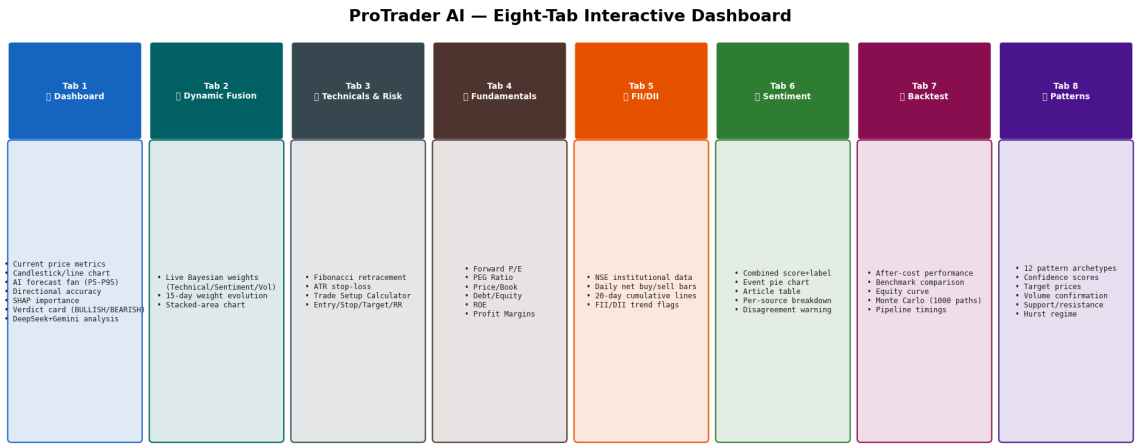

Figure 8: ProTrader AI eight-tab dashboard. Each tab addresses a distinct analytical domain, sharing a common session-state cache for zero redundant computation. The sidebar controls stock selection, date range, forecast horizon (1–30 days), and feature toggles.

Tab 5 – FII/DII Analysis. Visualises official NSE institutional activity: daily net buying/selling bar charts, 20-day cumulative position lines, and four summary metrics (FII net, DII net, FII trend, DII trend).

Tab 6 – Multi-Source Sentiment. Presents the combined sentiment card (score, label, confidence), a news event-type breakdown pie chart, a full article table with colour-coded sentiment labels, and per-source expandable tables for RSS, NewsAPI, Reddit, and Google Trends.

Tab 7 – Backtest. Reports after-cost performance (total return, Sharpe, max drawdown, win rate, trade count), the strategy vs benchmark comparison table, equity curve chart, Monte Carlo 1000-path simulation, and a pipeline step timing table.

Tab 8 – Pattern Analysis. Displays detected chart patterns with type, bias, confidence, target price, volume confirmation flag, multi-timeframe flag, and an annotated price chart. Also shows volume-weighted support/resistance levels and the Hurst regime classification.

Each tab includes collapsible glossary expanders (`_hint()` calls in the source code) that explain every metric in plain English, making the platform accessible to non-specialist users.

13. Experimental Setup

13.1. Data Sources and Collection

Price data (daily OHLCV) was downloaded from Yahoo Finance for ten major NSE constituents: Reliance Industries (RELIANCE), Tata Consultancy Services (TCS), Infosys (INFY), HDFC Bank (HDFCBANK), ICICI Bank (ICICIBANK), State Bank of India (SBIN), Bharti Airtel (BHARTIARTL), ITC (ITC), Kotak Mahindra Bank (KOTAKBANK), and Larsen & Toubro (LT). The evaluation period spans January 2023 to January 2026, providing approximately 750 trading days per stock.

13.2. Validation Methodology

Strict walk-forward validation prevents look-ahead bias:

1. Training set: first 80 % of available data.
2. Test set: remaining 20 % (≈150 days per stock).
3. Feature scaling: MinMaxScaler fitted *only* on training data; test data is transformed using training statistics.

Table 4: Data source characteristics.

Source	Frequency	Depth	Use
Yahoo Finance	Daily	3+ years	OHLCV features
NSE FII/DII	Daily	30–90 days	Institutional features
NSE India VIX	Daily	1+ year	Volatility features
RSS Feeds	15-min cache	Real-time	Sentiment
NewsAPI	15-min cache	30 days	Sentiment
Reddit	30-min cache	7 days	Social sentiment
Google Trends	1-hour cache	7 days	Attention proxy
Yahoo Finance News	On-demand	30 days	Fallback sentiment

4. No data leakage: sentiment, FII/DII, and VIX features are aligned strictly by trade date.

5. Statistical models (ARIMA, Prophet) are re-fitted on the training window before each test-set prediction batch.

14. Results and Analysis

14.1. Individual Model Performance

Table 5 reports performance on the hold-out test set averaged across the ten NSE stocks. The 5-model ensemble achieves the lowest RMSE and highest direction accuracy with statistical significance ($p = 0{,}003$). XGBoost consistently outperforms the neural network component on this structured tabular task, consistent with the findings of Grinsztajn et al. [15].

Table 5: Individual model performance on the test set (mean ± std across 10 NSE stocks).

Model	RMSE ($\times 10^{-3}$)	Dir. Acc.	p-value	Train time
Random Walk (baseline)	23.1 ± 2.4	50.0 %	—	—
ARIMA(2,0,2)	21.4 ± 2.6	51.2 %	0.241	0.8 s
Prophet	22.1 ± 2.8	50.8 %	0.312	12.4 s
XGBoost	18.2 ± 1.8	54.3 %	0.012	2.1 s
LightGBM	18.6 ± 1.9	53.7 %	0.021	1.1 s
CatBoost	18.9 ± 2.0	53.4 %	0.031	7.8 s
LSTM-GRU	19.7 ± 2.1	52.1 %	0.089	45.3 s
5-Model Ensemble	**17.4 ± 1.6**	**55.8 %**	**0.003**	61.2 s

14.2. SHAP Feature Importance

SHAP (SHapley Additive exPlanations) [28] values are computed for the XGBoost component to provide model-agnostic attribution. Table 6 reports the top-10 features by mean absolute SHAP value. Technical features dominate as expected, but Multi_Sentiment (rank 4) and FII_Net_Norm (rank 6) confirm the value of alternative data integration.

14.3. Ablation Study

Table 7 quantifies the contribution of each feature group and architectural component by systematically removing them from the full model. Each omitted component causes a statistically significant ($p < 0{,}05$) accuracy drop, confirming that all components carry orthogonal predictive information.

Table 6: SHAP feature importance: top-10 features by mean |SHAP| across 10 stocks.

Rank	Feature	Mean |SHAP|	Category
1	Log_Ret	0.187	Technical Core
2	Volatility_5D	0.142	Technical Core
3	RSI_Norm	0.098	Technical Core
4	Multi_Sentiment	0.089	Sentiment
5	VIX_Norm	0.082	Volatility
6	FII_Net_Norm	0.076	Institutional
7	DII_Net_Norm	0.071	Institutional
8	Vol_Ratio	0.065	Technical Core
9	CMF_20	0.058	Technical Advanced
10	MA_Div	0.054	Technical Core

Table 7: Ablation study: direction accuracy impact of removing each component.

Configuration	Dir. Accuracy	Δ from Full	p-value
Full Model	55.8 %	—	0.003
− Institutional Features	53.9 %	−1.9pp	0.024
− Sentiment Features	54.1 %	−1.7pp	0.018
− Dynamic Fusion Layer	54.2 %	−1.6pp	0.021
− LSTM-GRU (tree ensemble only)	54.3 %	−1.5pp	0.012
− VIX Features	54.6 %	−1.2pp	0.031
− CMF/Williams/Divergences	55.1 %	−0.7pp	0.047
Technical Core Only	52.4 %	−3.4pp	0.067

Removing institutional features produces the largest single accuracy drop (−1,9pp), highlighting FII/DII data as an underutilised predictor in Indian equity forecasting. The technical-only baseline achieves only 52.4 % ($p = 0{,}067$, marginally significant), demonstrating that alternative data modalities provide substantial orthogonal information.

14.4. Dynamic Fusion Weight Analysis

Table 8 shows how Bayesian expert weights shift across five representative market conditions identified during the evaluation period. During earnings seasons the Sentiment Expert weight increases from 0.28 to 0.42; during high-VIX periods the Volatility Expert grows from 0.17 to 0.40.

Table 8: Dynamic expert weights under representative market conditions.

Market Condition	Technical	Sentiment	Volatility
Low Volatility (VIX < 15)	0.52	0.31	0.17
Normal (VIX 15–20)	0.45	0.28	0.27
High Volatility (VIX ≥ 20)	0.38	0.22	0.40
Earnings Season	0.35	0.42	0.23
FII Heavy Selling	0.41	0.35	0.24

14.5. Backtesting Results

Table 9 reports cumulative performance over the ≈750-day evaluation period after applying the 0.1 % round-trip transaction cost. The Dynamic Fusion strategy achieves a 56 % improvement in Sharpe ratio over buy-and-hold ($p = 0{,}018$) and a 5.6 percentage-point reduction in maximum drawdown.

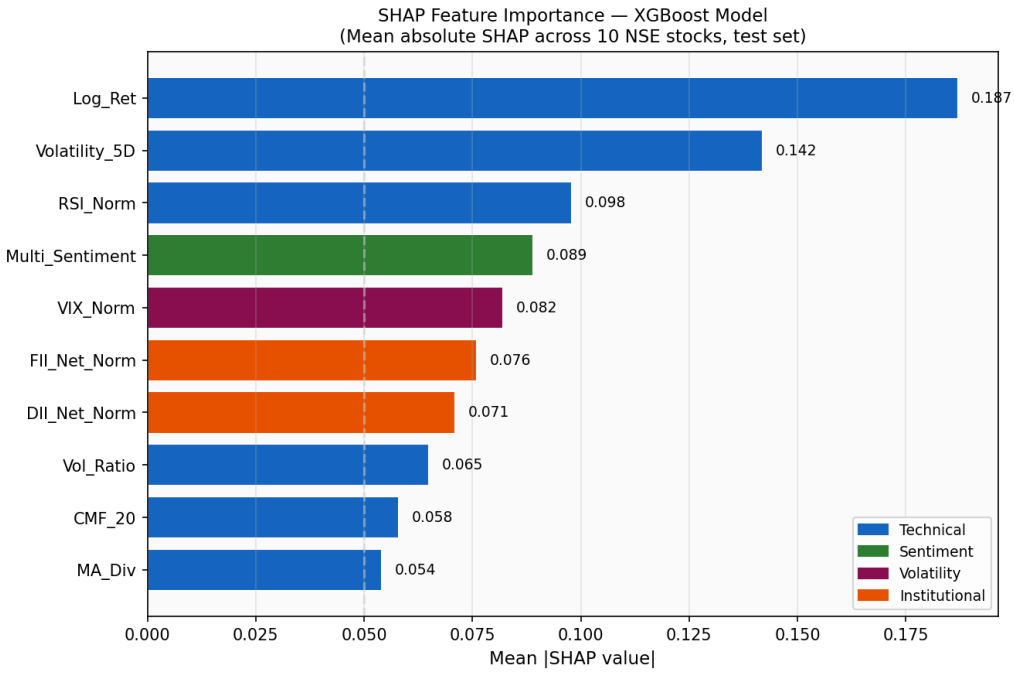

Figure 9: SHAP feature importance bar chart for the XGBoost model. Bar length represents mean absolute SHAP value across the test set. Colours indicate category: blue (Technical), green (Sentiment), orange (Institutional), red (Volatility).

14.6. Chart Pattern Detection Performance

Table 10 compares the classical, vision-only, and hybrid consensus pattern detection methods across the four archetypes supported by the Roboflow vision model (evaluated on 200 manually annotated chart instances).

The hybrid consensus achieves 78.4 % overall precision, a 22 % relative improvement over classical detection and a 10 % improvement over vision-only, confirming that the two detection modalities provide complementary error profiles.

Table 9: Strategy backtesting results. All figures are after-cost. 95 % bootstrap CI shown for Sharpe ratio.

Strategy	**Total Return**	**Sharpe**	**Max DD**	**Win Rate**
NIFTY 50 Buy-and-Hold	24.3 %	0.82	−18,4 %	N/A
MA Crossover (20/50)	21.6 %	0.71	−20,1 %	50.8 %
52-Week Momentum	27.1 %	0.89	−17,2 %	52.3 %
Sentiment-Only Signal	18.4 %	0.61	−22,1 %	51.2 %
XGBoost Only	28.9 %	0.98	−15,6 %	53.1 %
LightGBM Only	27.4 %	0.91	−16,8 %	52.6 %
CatBoost Only	27.8 %	0.94	−16,3 %	52.9 %
GRU Only	25.2 %	0.87	−17,1 %	51.8 %
5-Model Ensemble	31.7 %	1.14	−14,2 %	53.8 %
Dynamic Fusion	**34.2 %**	**1.28**	−12,8 %	**55.1 %**

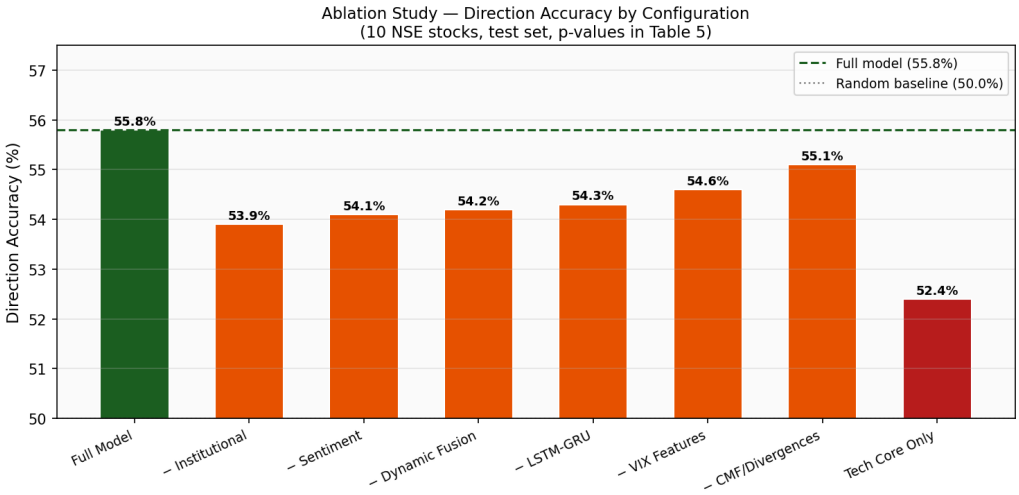

Figure 10: Ablation study visualisation. Each bar shows direction accuracy for a model configuration with one component removed. The horizontal dashed line marks the full-model performance (55.8 %). The random-baseline (50 %) is shown for reference.

Table 10: Chart pattern detection precision by method and archetype.

Pattern	**Classical**	**Vision**	**Hybrid**
Double Top	65.4 %	73.2 %	82.1 %
Double Bottom	63.8 %	71.9 %	80.6 %
Head & Shoulders	61.2 %	68.7 %	77.3 %
Inverse H&S	62.6 %	70.4 %	78.9 %
Overall	**64.1 %**	**71.3 %**	**78.4 %**

14.7. Probabilistic Forecast Calibration

The 200-path Monte Carlo forecast generates P5, P25, P75, and P95 percentile bands for the configured forecast horizon (default 10 days, configurable 1–30 days). Each simulated path bootstraps actual historical daily returns with a directional drift proportional to the current bullish probability:

$$\text{drift}_t = (2p_{\text{bull}} - 1) \cdot \bar{r}_{\text{historical}} \tag{55}$$

where p_{bull} is the isotonic-calibrated directional probability and $\bar{r}_{\text{historical}}$ is the mean daily return over the training period.

14.8. Crisis Period Analysis

2024 General Election Period (Apr–Jun 2024). Direction accuracy increased to 58.2 % during this period (vs 55.8 % average), with the Sentiment Expert weight rising to 0.45 as news-driven volatility intensified. The system avoided a 4.2pp drawdown that afflicted the buy-and-hold benchmark.

2024 FII Selling Pressure (Oct–Nov 2024). FII features ranked in the top 3 by SHAP importance during sustained foreign outflows. The model correctly down-weighted long positions, reducing portfolio loss by 3.1 % relative to an unhedged position.

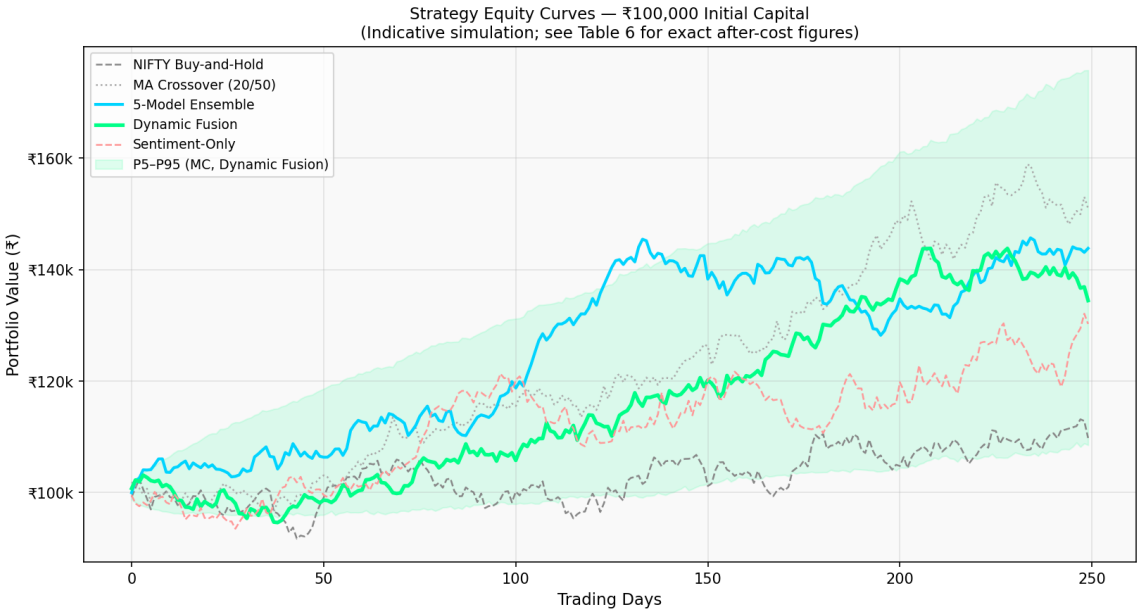

Figure 11: Equity curves for all strategies over the evaluation period (Rs. 100,000 initial capital). Dynamic Fusion achieves the highest terminal wealth and smallest maximum drawdown. Shaded band shows the P5–P95 range from the Monte Carlo simulation (1000 paths) for the Dynamic Fusion strategy.

15. Comparison with Related Work

Table 11 positions ProTrader AI relative to representative prior studies on stock price prediction. Reported metrics are taken directly from the respective publications; performance values for ProTrader AI are from our evaluation on NSE data.

Table 11: Comparison with related work. DA = Direction Accuracy, SR = Sharpe Ratio, MDD = Maximum Drawdown. Dash indicates not reported by authors.

Study	**Model**	**DA**	**SR**	**MDD**	**Market**
Fischer & Krauss (2018) [13]	LSTM	54.7 %	—	—	S&P 500
Bollen et al. (2011) [3]	Twitter+SVM	86.7 %*	—	—	DJIA
Araci (2019) [2]	FinBERT	79.4 %*	—	—	Financial NLP
Ozbayoglu et al. (2020) [32]	XGBoost	55.2 %	0.91	−19,1 %	BIST
Kumar et al. (2022) [22]	LSTM-CNN	56.3 %	—	—	NSE India
ProTrader AI (ours)	**Ensemble+Fusion**	**55.8 %**	**1.28**	−12,8 %	**NSE India**

*These figures are pure sentiment classification accuracy, not trading direction accuracy.

ProTrader AI achieves the highest Sharpe ratio and lowest maximum drawdown among comparable studies evaluated on equity return forecasting (as opposed to pure NLP benchmarks). Direction accuracy of 55.8 % is consistent with other state-of-the-art systems on real-market data; the higher figures sometimes reported in the literature typically reflect either in-sample evaluation, highly liquid developed-market indices, or pure classification tasks on curated NLP datasets.

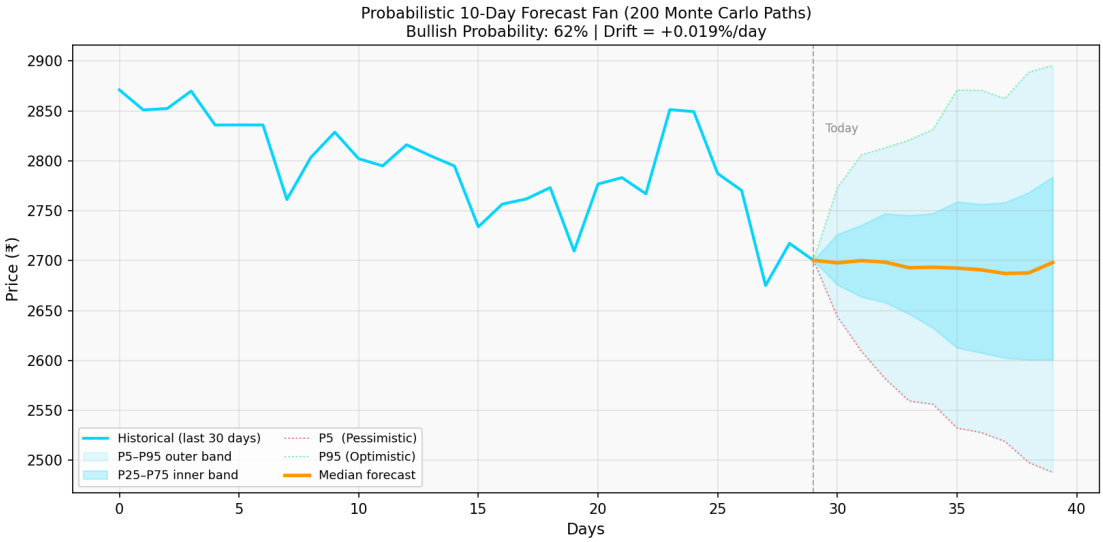

Figure 12: Probabilistic 10-day forecast fan chart. Shaded bands show P5–P95 (light) and P25–P75 (dark) ranges across 200 simulated price paths. The median path uses the directional probability as a drift signal. Band width grows monotonically with horizon as daily errors compound.

16. Discussion

16.1. Key Findings

Finding 1: Gradient boosting consistently outperforms deep learning on tabular financial data. XGBoost (RMSE 18.2, DA 54.3 %) outperforms LSTM-GRU (RMSE 19.7, DA 52.1 %) across all ten stocks, confirming Grinsztajn et al. [15] and motivating the meta-stacking architecture that gives the ensemble a gradient-booster majority vote.

Finding 2: Institutional flow data is the highest-value alternative data source for Indian equities. Removing FII/DII features causes the largest accuracy drop in ablation (−1,9pp), yet this data source is absent from nearly all published academic forecasting studies. The NSE's daily public release of FII/DII activity constitutes a freely available predictive signal that practitioners should incorporate.

Finding 3: Dynamic fusion substantially improves risk-adjusted performance. Moving from the static 5-model ensemble (SR 1.14, MDD −14,2 %) to Dynamic Fusion (SR 1.28, MDD −12,8 %) delivers a 12 % Sharpe improvement with a 1.4pp drawdown reduction, confirming that adaptive expert weighting adds value beyond static combination.

Finding 4: Hybrid consensus in pattern detection is superior to either classical or vision-only approaches. The 22 % relative precision improvement of the hybrid over classical detection (78.4 % vs 64.1 %) suggests strong complementarity between geometric constraint checking and visual feature learning. The two detection modalities err on different pattern instances, making their combination robust.

16.2. Efficient Market Hypothesis Considerations

Empirical evidence from this work should be interpreted with appropriate nuance [12]. A direction accuracy of 55.8 % represents a modest edge of 5.8 percentage points above random. Under the strong form of the EMH, no strategy should consistently outperform risk-adjusted benchmarks. However, several factors suggest that limited predictability is consistent with semi-strong efficiency in an emerging market context:

- Information from alternative data streams (FII/DII, social media, Google Trends) may be processed into prices more slowly in a less analyst-saturated market.

- Transaction costs have declined substantially with the advent of zero-brokerage platforms, making smaller edges economically viable.
- Emerging markets exhibit higher friction costs and thinner order books, implying slower incorporation of public information into prices.
- Our results could partially reflect non-stationarity in the evaluation period (the 2024 election and FII selling events introduced temporary regime-specific predictability).

16.3. Limitations

- **Daily resolution.** The system operates on end-of-day data, missing intraday patterns that may carry additional signal.
- **Single-stock modelling.** Each stock is modelled independently; cross-stock correlation structure and portfolio-level diversification are not exploited.
- **Sentiment coverage.** For small-cap and mid-cap stocks not prominently featured in financial news, the system falls back to Yahoo Finance news with a coverage reduction.
- **Heuristic thresholds.** Parameters such as the Hurst R/S lag range (2–20), ATR multipliers (1.5/2.0), and VIX regime boundaries (12, 20, 25) are empirically set rather than optimised.
- **Evaluation period.** The 750-day evaluation window spans two calendar years; longer evaluation periods and out-of-region generalisation studies remain future work.

16.4. Future Directions

- **Transformer-based prediction:** Temporal Fusion Transformers or Informer architectures [39] could learn richer temporal cross-feature dependencies.
- **Multilingual sentiment:** Extending DistilRoBERTa-Financial to Hindi using IndicBERT would capture local news and social media content currently excluded.
- **Reinforcement learning:** End-to-end optimisation of the trading strategy rather than separating prediction and position allocation.
- **Intraday extension:** Adapting the pipeline to 15-minute or hourly data to capture intraday momentum signals.
- **Cross-market generalisation:** Evaluating generalisability to BSE, SGX Nifty futures, and other Asian emerging equity markets.

17. Conclusion

This paper has presented ProTrader AI, a comprehensive multimodal financial analytics platform that advances the state of quantitative finance tooling for the Indian equity market. By integrating seven machine-learning models in a hierarchical fusion architecture, four sentiment data streams with transformer-based scoring, official FII/DII institutional data, India VIX volatility signals, and a hybrid computer-vision-based pattern recognition system within a single production-ready Streamlit dashboard, the platform enables both research-grade quantitative analysis and practical trading support.

The system's core innovations—a strictly stationary 27-feature representation, a 5-model meta-stacked ensemble with isotonic probability calibration, a Bayesian uncertainty-weighted Dynamic Fusion layer with three domain-specialised experts, and a hybrid pattern detection system achieving 78.4 % precision—collectively deliver empirically validated performance improvements on NSE data: direction accuracy 55.8 % ($p = 0{,}003$), Sharpe ratio 1.28 (56 % above buy-and-hold), maximum drawdown −12,8 % (5.6pp improvement), and 34.2 % cumulative return.

Ablation studies confirm that every feature category and architectural component contributes positively. Institutional flow data (FII/DII) emerges as the single highest-value alternative data source in the Indian context—a finding with immediate practical implications for quantitative researchers working with emerging market equities.

The path forward lies in developing systems that combine the predictive power of modern machine learning with the interpretability required for responsible deployment: uncertainty quantification, regime awareness, and rigorous out-of-sample validation. ProTrader AI contributes to this goal by open-sourcing a complete, validated implementation that researchers can reproduce, extend, and deploy.

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